

```

import matplotlib.pyplot as plt

# Function to find the largest rectangle area
def largestRectangleArea(heights):
    stack = []
    max_area = 0
    left = right = height = 0

    h = heights + [0] # Add a dummy bar

    for i in range(len(h)):
        while stack and h[i] < h[stack[-1]]:
            top = stack.pop()
            current_height = h[top]

            if stack:
                width = i - stack[-1] - 1
                start = stack[-1] + 1
            else:
                width = i
                start = 0

            area = current_height * width

            if area > max_area:
                max_area = area
                left = start
                right = i - 1
                height = current_height

```

```
stack.append(i)
```

```
return max_area, left, right, height
```

```
# ----- MAIN PROGRAM -----
```

```
# Input
```

```
n = int(input("Enter number of bars: "))
```

```
heights = []
```

```
print("Enter the heights of the bars:")
```

```
for i in range(n):
```

```
    heights.append(int(input()))
```

```
# Find largest rectangle
```

```
area, left, right, rect_height = largestRectangleArea(heights)
```

```
print("\nHeights =", heights)
```

```
print("Largest Rectangle Area =", area)
```

```
# Draw histogram
```

```
plt.figure(figsize=(8,5))
```

```
for i in range(len(heights)):
```

```
    if left <= i <= right:
```

```
        plt.bar(i, heights[i], color="red", edgecolor="black")
```

```
    else:
```

```
        plt.bar(i, heights[i], color="skyblue", edgecolor="black")
```

```
# Draw rectangle boundary
```

```
plt.plot(  
    [left-0.4, right+0.4, right+0.4, left-0.4, left-0.4],  
    [rect_height, rect_height, 0, 0, rect_height],  
    color="black",  
    linewidth=2  
)
```

```
plt.title("Histogram with Largest Rectangle")
```

```
plt.xlabel("Bar Number")
```

```
plt.ylabel("Height")
```

```
plt.grid(axis='y', linestyle='--', alpha=0.5)
```

```
plt.show()
```

